

Weather Wise AI

Phase 6: Project Testing

Testing verifies that Weather Wise AI meets the requirements and works reliably across devices and conditions.

Types of Testing

- Unit testing: individual functions (data cleaning, API parsers, model prediction).
- Integration testing: frontend, backend, database and external API working together.
- System testing: complete user journeys from search to alert.
- Performance testing: response time and behaviour under multiple users.
- Model evaluation: MAE/RMSE versus the baseline on unseen data.
- Usability testing: feedback from a few real users.

Sample Test Cases

ID	Test	Expected result	Status
TC01	Search valid city	Current weather displayed	Pass
TC02	Search invalid city	Friendly error message	Pass
TC03	Ask 'Will it rain today?'	Correct, clear AI answer	Pass
TC04	Weather API unavailable	Cached data shown with notice	Pass
TC05	Load on mobile screen	Layout adapts correctly	Pass
TC06	Alert condition triggered	User receives notification	Pass

Testing Notes

- Update the Status column with your actual results and record any bugs with steps to reproduce.
- Re-test after every fix (regression testing) before the final demo.

